is more likely to be incorrectly marked as violating content than others.'

### 3.1 Speech suppression measures for Boolean flags

We begin by defining speech measures in the context of Boolean-flag content moderation outputs, where a true (1) return value indicates violating content. Let $Y$ denote dataset labels, $\hat{Y}$ denote predicted labels (i.e., the resulting content moderation flags), and $I$ denote the set of identity groups. In line with the definitions discussed in [24, 27], we define a *per-identity-group false positive rate* ($i$-FPR) as follows:

Definition 1 ($i$-FPR). $P[\hat{Y} = 1 | Y = 0, I = i]$

Speech suppression is then quantified by taking the ratio between the overall (full dataset) FPR and the $i$-FPR. It's common for fairness measures to be defined such that the optimization goal is a ratio with a value of 1.0 or a difference with a value of 0.0, and previous work on bias in content moderation has used the difference [24]. Following recent work that demonstrates that the ratio is more appropriate for most fairness contexts due in part to its ability to highlight disparities even with small numerical values [71], we define speech suppression accordingly:

Definition 2 ($i$-Speech Suppression). $\frac{i\text{-}FPR}{FPR}$

Values of 1.0 show that the identity group FPR is the same as the overall rate while values more than 1.0 indicate that the identity group's FPR is worse than the overall rate, i.e. the group suffers from identity-related speech suppression by having text incorrectly flagged more often than the base rate for the dataset. In addition to comparison versus results on the dataset overall, normalizing based on the overall FPR allows comparison across APIs regardless of the API's false positive rate. In cases where we are interested in determining the identity group with the worst speech suppression, we will additionally consider: $\max_{i \in I}$ $i$-speech suppression.

### 3.2 Speech suppression measures for numerical scores

Some content moderation APIs return scores (overall and/or per-category) instead of, or in addition to, Boolean flags. Scores are positive and structured so that higher scores indicate content identified as more likely to violate the guidelines. Per-category scores could indicate, e.g., that text scores high for moderation policies on violence but not on sexual content. Following a per-identity approach similar to that for Boolean flags, we consider per-identity average (median) scores. Given our focus on incorrectly flagged non-violating speech, we consider averages solely over the true negative instances. This choice allows us to avoid needing to adjust scores in the case where some identity group has more truly violating speech associated with related text, for example in the likely case where a marginalized group has more hate speech associated with identity-related text. Considering only the true negative instances allows us to expect that the content moderation scores will be low across identity groups.

Formally, consider a dataset $(\mathbb{X}, \mathbb{I}, Y)$ with data instance $j$ including data $X_j \in \mathbb{X}$, identities $I_j \in \mathbb{I}$, and label $y_j \in Y$. Let $s_c(X_j)$ denote the resulting content moderation score on instance $X_j$ for category $c \in C$. Let $X^{i,0} = \{X_j \in \mathbb{X} \mid i \in I_j, y_j = 0\}$ be the subset of data instances associated with identity group $i \in I$ that should not be flagged as content violations (where true label $y_j = 0$). In cases where a content moderation API returns multiple category scores, we associate each instance with its maximum score – the score most likely to cause the content to be flagged. We calculate the median over all true negative instances per identity group:

Definition 3 ($i$-median).

$$median\left(\left\{\max_{c \in C} s_c(X_j) \mid X_j \in X^{i,0}\right\}\right)$$

As before, we create a speech suppression score from this value by taking the ratio between the per-identity group median and the median of worst instance scores for the true negative values:

Definition 4 ($i$-speech suppression).

$$\frac{i\text{-}median}{median\left(\left\{\max_{c \in C} s_c(X_j) \mid y_j = 0\right\}\right)}$$

When interested in the identity group with the highest (worst) scores, we will take: $\max_{i \in I}$ $i$-speech suppression.

## 4 AUDIT METHODOLOGY

In order to assess the identity-related speech suppression due to automated content moderation systems across differing types of content, identities, and APIs, we develop an audit methodology based on three key steps:

(1) content moderation texts from both traditional online moderation and generative AI contexts, with labels indicating if content should be allowed;
(2) identity categorization of each text as associated with each of nine identities: men, women, christian, non-christian, straight, LGBT, white, non-white, or disability; and
(3) running each of five publicly available content moderation APIs.

An overview of the audit methodology can be found in Figure 1 and data and code to replicate these experiments can be found at: https://github.com/genAIaudits/speech-suppression.

### 4.1 Datasets

We build an automated content moderation benchmark from seven datasets; five from previous work and two introduced in this work. The five datasets from previous work were chosen because of their previous use assessing content moderation systems (see, e.g., [48]). These previous datasets include Boolean classification labels indicating whether the provided text snippet should be categorized as hateful, toxic, or otherwise offensive as determined by human raters. As we will discuss below, we introduce a new methodology for creating such labels in the context of speech suppression that takes advantage of its focus on incorrect flagging behavior, and use this methodology to build on existing human-curated information to label the two new data sets.

*Traditional datasets.* Four of the included datasets were created to help automatically monitor online comment sections and other short-form content (e.g., tweets), and are generally reflective of

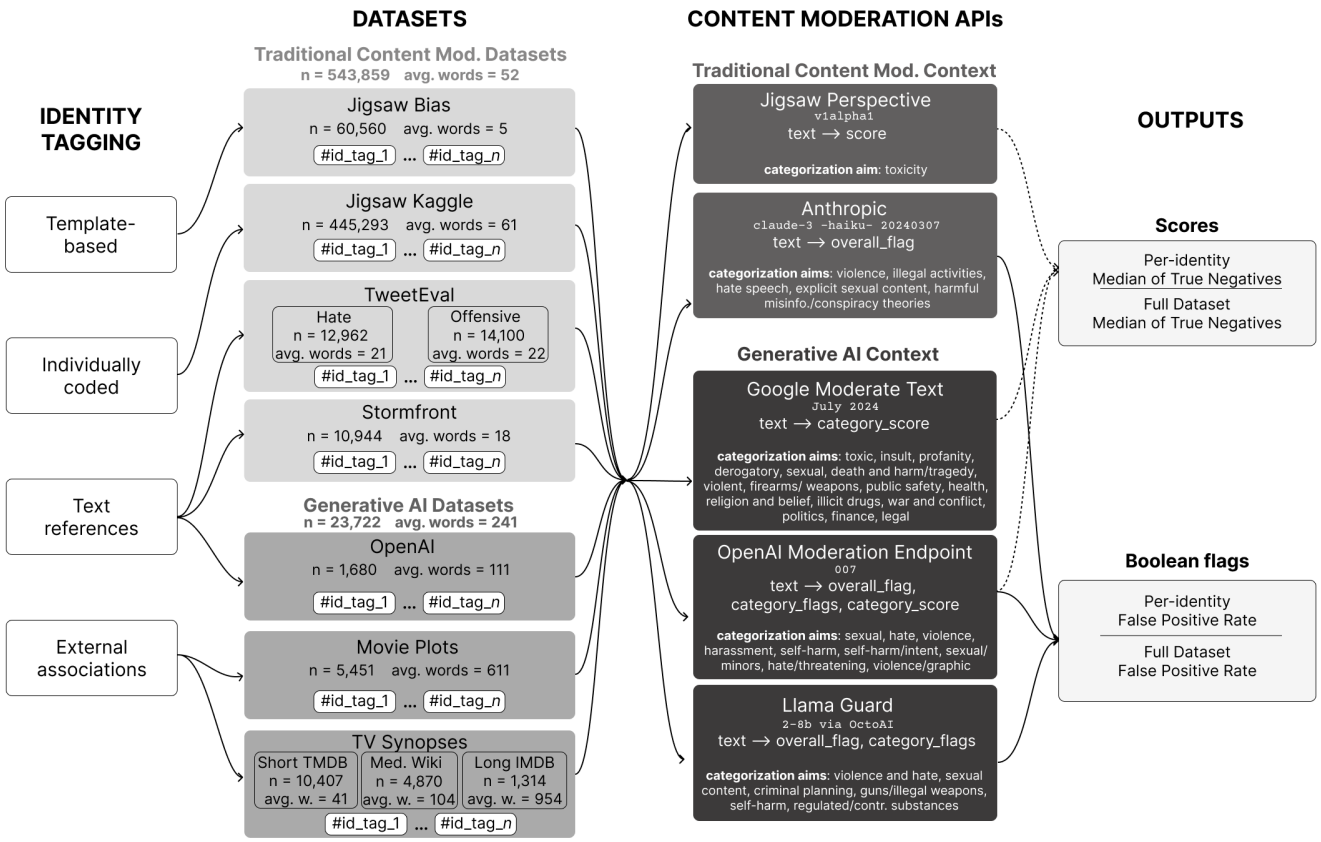


**Figure 1: Overview of the audit methodology pipeline. Dataset information shown includes total number of instances ($n$) and average number of words per instance. Content moderation API information shown includes the version number audited, return types, and categorization aims.**

these goals. These are the Jigsaw Kaggle [14, 20], Jigsaw Bias [24], Stormfront [22], and TweetEval [8] datasets. The Jigsaw Bias and Stormfront dataset labels indicate whether the text is toxic, the TweetEval dataset provides a subset labeled to indicate hate and another labeled for offensive content, and the Jigsaw Kaggle dataset provides content categorized based on an overall toxicity flag and six harmful content types (severe toxic, obscene, threat, insult, identity hate, and sexually explicit). All these datasets contain text items that are fairly short, with an average length between 5 and 61 words. Jigsaw Kaggle's dataset includes about 2 million text instances; we include only the subset of 445, 293 that were also manually identified for association with an identity group. See Figure 1 for dataset summary statistics; when considered as a group, we identify all these datasets (Jigsaw Kaggle, Jigsaw Bias, Stormfront, and TweetEval) as "Traditional" content moderation datasets.

*Generative AI data.* More recent datasets have been developed to test generative AI content moderation systems. This includes OpenAI's content moderation dataset [48] we include in our benchmark that contains longer-form text tagged with information about whether it should be categorized as any of nine types of violating content (sexual, hate, violence, harassment, self-harm, self-harm with intent, sexual content relating to minors, threatening hate speech, or graphic violence). [46] also develop a dataset of television episode synopses, however the dataset is fairly small (1,392 episodes); inspired by their focus on cultural content, we introduce additional datasets containing television and movie plots.

*4.1.1 New datasets: Television and movie plots.* In order to test longer-form creative content of the type relevant to generative text systems and our associated concerns about speech suppression, we developed datasets of television episode and movie plot synopses. Using The Movie Database (TMDB)[65], we gathered the top 10, 000 television shows and movies as of Summer 2024 and filtered the lists to only include English-language shows and movies released in the United States. For each television show, overviews for all episodes in the first season were collected. The show and movie names were then identified on Wikipedia, where episode synopses and movie plots were collected. Longer user-generated summaries of TV episodes were additionally collected from IMDB. The summary statistics for the OpenAI, Movie Plots, and TV synopses data, collectively referred to as the GenAI datasets, are also given in Figure 1.

It can be difficult for researchers outside of an industry context to create new datasets for content moderation tasks, since many of the previously developed traditional datasets were hand-labeled by contract or "gig" workers and this may be cost prohibitive for the development large datasets in an academic context. Our focus on speech suppression allows a wider range of collected data to be useful to our task; as we describe further in what follows, we were able to automatically generate labels based on television or movie age ratings, taking advantage of previous human labor applicable to our goals.

We collected age ratings from TMDB for movies (G, PG, PG-13, R, and NC-17) and from IMDB for television episodes (TV-Y, TV-Y7, TV-G, TV-PG, TV-14, TV-MA). These age ratings are established by an external organization (the Motion Picture Association) and indicate the maturity level of the content; for example, PG-13 content is considered appropriate for people aged 13 and up. Using these age ratings, we constructed two sets of labels for each TV episode or movie: `PG-ok` and `PG-13-ok`. The `PG-ok` labels classify an episode or movie as appropriate, or not in need of content moderation, if it's rated G or PG (respectively for TV shows, TV-Y, TV-Y7, TV-G, or TV-PG), and the `PG-13-ok` labels mark G, PG, or PG-13 (for TV shows, TV-Y, TV-Y7, TV-G, TV-PG, or TV-14) rated episodes or movies as appropriate. Unrated episodes or movies were excluded from the final dataset. Additionally, since PG-13 ratings were not introduced for movies until July, 1984 and age ratings were not introduced for television shows until January, 1997, we include only movies released after 1985 and television shows released after 1997. We will report detailed results on both `PG-ok` and `PG-13-ok` labels, and where aggregating across datasets will use the `PG-13-ok` labels for these datasets.

These introduced age-based labels align well with our speech suppression measurement goals, since they focus on incorrectly flagging content as violating, and would be less appropriate to use to identify content that should be flagged. Since we include only episode and movie synopses in our data, and not, e.g., the full scripts, there is likely to be content missing from the synopses that is included in the full episode or movie as rated. Under a conservative interpretation of these ratings for our `PG-13-ok` labels, the movie synopses for PG-13 rated movies are certainly appropriate for kids 13 and older (i.e., should not be flagged), and higher rated movies may also have appropriate synopses since TMDB, Wikipedia, and IMDB are unlikely to contain inappropriate text even for shows and movies with higher age ratings. We note that using the `PG-13-ok` labels makes sense for all AI services, since they include a requirement in their terms of use that users must be at least 13 years old [56], or in some cases at least 18 years old [4], likely to comply with COPPA, a U.S. law that imposes additional restrictions on websites directed to children under 13 [67].

## 4.2 Identity categorization

In support of our goal to assess identity-related bias in speech suppression, we need each text to be associated with zero, one, or multiple identity groups. Specifically, we aimed to determine if each text instance should be associated with identity categories based on race, gender, religion, sexual orientation, and disability. The choice of these identity categories is motivated by U.S. non-discrimination law (they are protected categories under civil rights laws and interpretations) and are common choices in the literature, including the literature and datasets we directly build upon (see, e.g., [14, 24, 46]). In what follows, we will group identities to compare dominant versus non-dominant groups, as is common practice in quantitative work for handling marginalized groups' small sample sizes, and qualitative work comparing dominant group outcomes to others (see e.g., [29, 30, 50]).

Only two of the datasets in our benchmark were previously tagged with identity information; the Jigsaw Bias and Jigsaw Kaggle datasets. The Jigsaw Bias dataset [24] is *template-based*, substituting identity terms into various repeated phrases (e.g., "hug gay", "hug male", and so on) so we can directly extract those identity groups. The Jigsaw Kaggle dataset includes human *individually coded* identity labels [14, 20] for a subset of about 445K instances; we directly use these individually coded identities.

The remaining five datasets do not already include identity attributes, so we create these associations following two main strategies: identification of explicit *text references* to an identity group, and *external association* of cultural content with an identity group. For both introduced methodologies, we follow the strategy of associating text with *specific* identities (e.g., Black or lesbian) and then grouping those identities into *general* identity groups (e.g., non-white or LGBT) in order to create groups large enough to assess content moderation trends. This strategy of going from specific identities to larger more general identity groups allows us to categorize text according to fine-grained information (e.g., a slur specifically used about lesbians) while still considering trends over a larger sized group.

*4.2.1 Text references.* In order to automatically tag text as related to an identity group, we determined that we would need a wide variety of terms describing identity groups to draw from. Once we have an appropriate list of such terms, we can then tag any text that explicitly references a group as related to that identity. We turned to Wikipedia for pages identifying slurs and slang terms associated with various identity groups. We believe that Wikipedia's wide range of editors from a variety of backgrounds and its consensus-building joint editorial process make it useful for building out identity-related information, such as these lists of slurs or slang terms. Related work has found that Wikipedia content on taboo topics is of relatively higher quality, and that community-driven Wikipedia curation can be useful in the context of AI evaluation [18, 41].

We created lists of identity-related terms that include both slurs or slang (Appx. Tables 6 and 7) and neutral descriptors (Appx. Table 8) about an identity group. These lists were collected from Wikipedia (e.g., Wikipedia's "List of ethnic slurs"), and then curated to remove terms that are also commonly used words in other contexts (e.g., "black") so as to conservatively identity-tag text as referring to a given identity group. Text from the Stormfront, TweetEval, and OpenAI datasets were then categorized as associated with a specific identity if they included any of these identity-related terms.

*Validation.* We were able to validate this identity categorization scheme using the Jigsaw Kaggle data's manually identified labels,